

Bagging Classifier

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Where this fits: Pipeline map steps 8 (Model), 9 (Evaluate), 10 (Tune). See the [Course map](#).



- **Builds on:** Simple imputation (mean, median, mode, constant) ([Note 37](#)); ML pipelines ([Note 38](#)); Bias-variance trade-off ([Note 62](#)); Overfitting ([Note 91](#)); Hyperparameter tuning ([Note 98](#)); Decision trees ([Note 100](#)).
- **Leads to:** Random forest ([Note 108](#)).
- **Compare with:** Boosting ([Note 101](#)); Voting ensembles ([Note 104](#)); Cross-validation ([Note 104](#)); Random forest ([Note 108](#)); Optuna ([Note 134](#)); Bayesian optimisation ([Note 134](#)).

1. Overview

Key point: scikit-learn's `BaggingClassifier` trains many copies of one classifier on random samples of the rows and/or columns and takes their majority vote. Its settings choose the type: bagging, pasting, random subspaces or random patches.

The idea was explained in the [bagging Note](#): bootstrapping, then aggregation, in four variants. This Note applies it to classification:

- a demo of decision surfaces, comparing one model with its bagged version, for each of the four types;
- `BaggingClassifier` in code on a dataset of 10,000 rows, with its hyperparameters;
- the out-of-bag score;
- what works in practice, and tuning with `GridSearchCV`.

The Notebook (`notebook.ipynb`) runs every experiment. The Dash app `app.py` has a control for every bagging setting and redraws the decision surfaces.

2. Seeing bagging on decision surfaces

Key point: On the moons data, one fully grown tree overfits (test accuracy 0.856); bagging 100 trees smooths the boundary and reaches 0.912.

The demo data is the two-moons toy dataset (500 points, 375 for training), the same as in the [decision tree hyperparameters Note](#), section 4. The app asks for six settings, the main hyperparameters of `BaggingClassifier`:

- **base model (estimator):** decision tree (the default), KNN or SVM;
- **n_estimators:** how many base models;
- **max_samples:** how many rows each model gets (here out of 375);
- **bootstrap:** draw those rows with replacement (`True`) or without (`False`);
- **max_features:** how many columns each model gets (here out of 2);

- **bootstrap_features:** draw those columns with replacement or without.

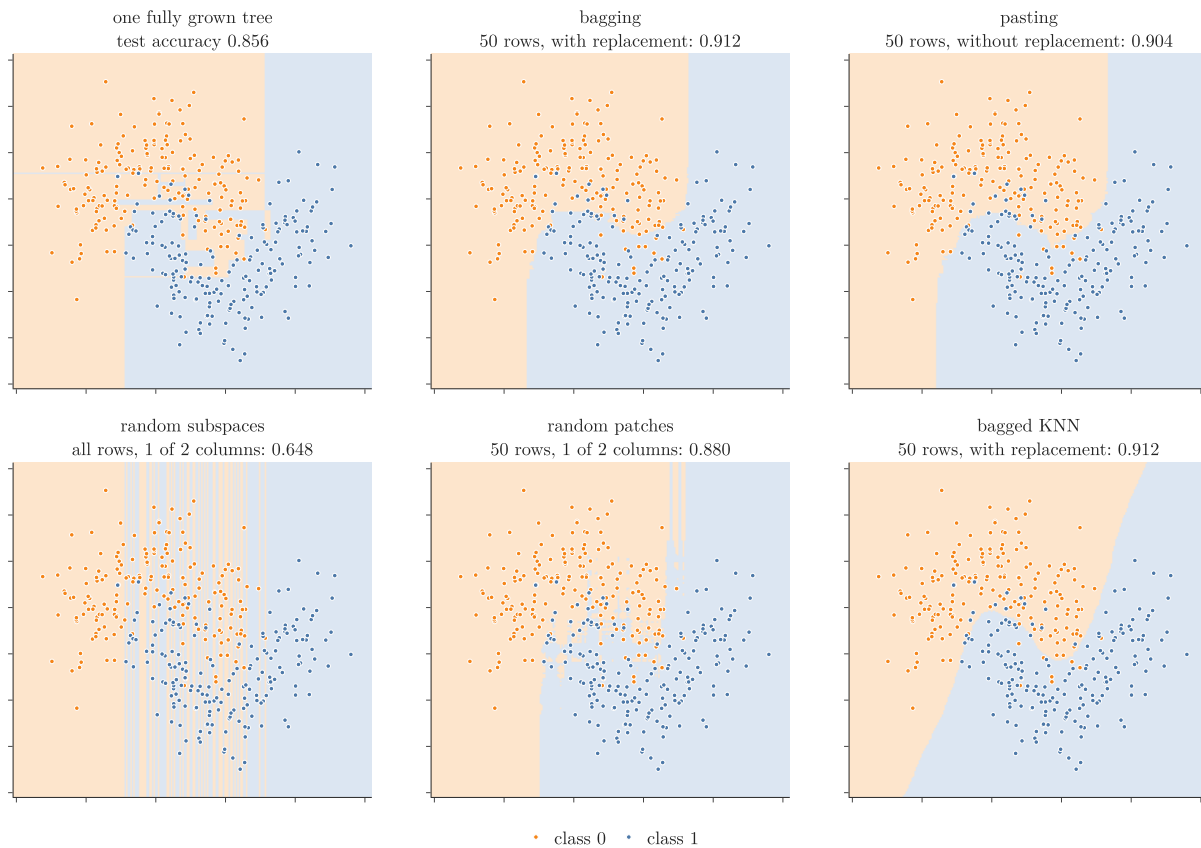


Figure 1: Decision surfaces on the moons data: one fully grown tree, then 100-tree ensembles of each bagging type, and bagged KNN. Titles give the test accuracy

2.1 Bagging against a single tree

Key point: The tree draws small boxes around single points; the bagged trees draw one smooth boundary.

With a decision tree, 100 estimators, 50 rows each, drawn with replacement, and both columns:

- **One tree** (Figure 1, top left): 0.856. Its surface has small boxes of one colour inside the other: overfitting. It is right on the training data, wrong on new data.
- **Bagging** (top middle): 0.912. The boundary is much smoother: the bias stays low and the variance falls, as the [bagging Note](#) predicted.

2.2 Other base models

Key point: Bagging KNN gives a slightly smoother boundary; bagging SVMs makes them a little worse.

Any classifier can be bagged:

- **KNN:** a single KNN ($k = 5$) already gives a smooth boundary and scores 0.912. Bagged, its boundary is a little smoother still (Figure 1, bottom right), with the same 0.912.
- **SVM:** a single SVM scores 0.896, but 100 bagged SVMs score only 0.872.

In practice, decision trees are bagged most often, KNN sometimes, SVMs rarely.

2.3 More estimators, more rows

Key point: Beyond a point, more models or more rows change little.

With 500 trees instead of 100 (50 rows each), accuracy moves only from 0.912 to 0.920. With 500 trees of 150 rows each it is 0.904. More estimators usually help up to some number, then stop making a difference. Both settings are worth tuning.

2.4 Pasting, random subspaces and random patches

Key point: Pasting behaves like bagging here; sampling one of only two columns hurts badly.

- **Pasting** (50 rows, `bootstrap=False`): 0.904. On a small toy dataset there is little difference from bagging; on large real datasets the difference can show.
- **Random subspaces** (all 375 rows, `bootstrap=False`, `max_features=1`, `bootstrap_features=True`): **0.648**, much worse than one tree. Each tree sees only one column, so it can only cut along one axis. The surface becomes vertical stripes (Figure 1, bottom left).
- **Random patches** (50 rows with replacement, 1 column): 0.880. Row sampling brings back some variety, but one column is still too little.

Column sampling makes sense only when there are many columns: 10, 20, 50, 100 or more. With two columns, sample rows (bagging or pasting) instead.

3. BaggingClassifier in code

Key point: On 10,000 rows with 10 columns, one tree scores 0.927; 500 bagged trees score 0.945.

3.1 The data and a single tree

Key point: `make_classification` builds 10,000 rows; 8,000 train, 2,000 test.

We make a dataset of 10,000 rows and 10 columns with `make_classification` (the [perceptron code Note](#)), of which 3 columns carry the information. 8,000 rows go to training and 2,000 to testing. A single fully grown decision tree scores **0.9265**: the number to beat.

3.2 Bagging

Key point: 500 trees, each trained on 2,000 rows (25%) drawn with replacement.

Python: A bagging classifier.

```
from sklearn.ensemble import BaggingClassifier
from sklearn.tree import DecisionTreeClassifier
```

```
bag = BaggingClassifier(
    estimator=DecisionTreeClassifier(),
    n_estimators=500,
    max_samples=0.25,  # 25% of 8,000 rows
    bootstrap=True,    # with replacement
    random_state=42,
    n_jobs=-1)         # all CPU cores
bag.fit(X_train, y_train)
accuracy_score(y_test, bag.predict(X_test))  # 0.945
max_samples is a share of the rows when it is a decimal (0.25 means 25%) and a row
count when it is a whole number (50 means 50 rows). max_features works the same
way for columns.
```

Bagging scores **0.945**, against 0.9265 for one tree. Training takes a while: 500 trees are trained instead of one. `n_jobs=-1` spreads them over every CPU core.

Extra: Older code passes the base model as `base_estimator=`. That name was deprecated in scikit-learn 1.2 and removed in 1.4; the current name is `estimator` (scikit-learn 1.2 changelog). Also, `max_samples` now defaults to `None`, which means “as many rows as the training set”, the same as the old default of 1.0.

3.3 Which rows and columns each tree got

Key point: Two attributes record what each tree saw: `estimators_samples_` (its rows) and `estimators_features_` (its columns).

After training, the classifier records what each base model saw:

- `bag.estimators_samples_[0]` holds the row numbers the first tree was trained on: 2,000 of them (25% of 8,000), starting 2523, 3113, 7114, ... Rows can repeat, because of replacement.
- `bag.estimators_features_[0]` holds its columns: all 10, `[0 1 2 ... 9]`, because we did not sample columns.

With `max_samples=0.5`, each tree gets 4,000 rows instead, and the accuracy is 0.950.

3.4 Other base models, pasting, subspaces and patches

Key point: Bagged SVMs 0.913, pasting 0.946, random subspaces 0.942, random patches 0.938: all within a point or so of bagging, except the SVMs.

Every variant is the same class with different settings:

Variant	Settings (all with 500 estimators)	Test accuracy
Bagging	<code>max_samples=0.25,</code> <code>bootstrap=True</code>	0.945
Bagged SVMs	<code>estimator=SVC()</code> , otherwise as bagging	0.9125
Pasting	<code>max_samples=0.25,</code> <code>bootstrap=False</code>	0.946
Random subspaces	<code>max_samples=1.0,</code> <code>bootstrap=False,</code> <code>max_features=0.5,</code> <code>bootstrap_features=True</code>	0.9415
Random patches	<code>max_samples=0.25,</code> <code>bootstrap=True,</code> <code>max_features=0.5,</code> <code>bootstrap_features=True</code>	0.938

- The **SVMs** do worse than trees and take much longer, one reason trees are the usual choice.
- For **pasting**, we also set `verbose=1`, which prints training progress, and `n_jobs=-1`.
- For **random subspaces**, `estimators_samples_[0]` now has all 8,000 rows, and `estimators_features_[0]` has 5 columns, for example `[9 2 9 7 7]`. Column 9 and column 7 appear twice because `bootstrap_features=True` draws columns with replacement.

4. The out-of-bag score

Key point: About 37% of the rows are never drawn for a given tree; scoring each tree on its unseen rows gives a free estimate of test accuracy.

The rows a tree never drew, about 37% of them (the [bagging Note](#), section 2.3), are its out-of-bag (OOB) rows. With `oob_score=True` (it needs `bootstrap=True`), scikit-learn scores the ensemble on them, so no separate test set is needed:

Python: The out-of-bag score.

```
bag = BaggingClassifier(DecisionTreeClassifier(),
                        n_estimators=500, max_samples=0.25,
                        bootstrap=True, oob_score=True,
```

```

        random_state=42, n_jobs=-1)
bag.fit(X_train, y_train)
bag.oob_score_      # 0.9429

```

The OOB score, **0.943**, is close to the real test accuracy, **0.945**. The [OOB score Note](#) explains how it is computed and when it can be trusted.

5. What works in practice

Key point: Bagging usually beats pasting; start `max_samples` at 0.25 to 0.5; sample columns only with many columns; tune with `GridSearchCV`.

From experiments on many datasets:

1. **Bagging usually beats pasting.** Drawing with replacement makes the samples more varied, so the ensemble's variance is lower and it does better on new data, at the cost of a little more bias. Since it is only `bootstrap=True` or `False`, try both.

The theory agrees: the variance of an average falls as the models become less correlated (ESL §15.2), and the Notebook's last cell confirms it on the sine data of the [bagging Note](#), section 3.2: changing only `bootstrap`, bagging's trees are less correlated (0.82 against 0.84), its variance lower (0.041 against 0.055) and its squared bias a little higher (0.0019 against 0.0013).

2. **`max_samples` between 0.25 and 0.5** usually works best for row sampling. Start at 0.25.
3. **Column sampling** (random subspaces, random patches) is for **high-dimensional** data, with many columns. With few columns it hurts, as Figure 1 showed.
4. **Tune with `GridSearchCV` or `RandomizedSearchCV`** instead of guessing (the [KNN Note](#), section 4.2, and the [regression trees Note](#), section 7.2).

Python: Grid search over the bagging settings.

```

parameters = {"n_estimators": [50, 100, 500],
              "max_samples": [0.1, 0.4, 0.7, 1.0],
              "bootstrap": [True, False],
              "max_features": [0.1, 0.4, 0.7, 1.0]}
search = GridSearchCV(BaggingClassifier(random_state=42),
                      parameters, cv=5, n_jobs=-1)
search.fit(X_train, y_train)
search.best_params_

```

This tries $3 \times 4 \times 2 \times 4 = 96$ combinations with 5-fold cross-validation: 480 fits, which takes several minutes (about 12 minutes on our 12-core machine). The best: **500 trees, 70% of the rows without replacement (pasting), 70% of the columns**. Its cross-validation accuracy is 0.955, and its test accuracy 0.952, the best of all our settings.

Here pasting wins: the rules are a starting point; the search decides.

6. Summary

Hyperparameter	Default	Meaning
<code>estimator</code>	None (decision tree)	the base model (formerly <code>base_estimator</code>)
<code>n_estimators</code>	10	number of base models
<code>max_samples</code>	None (all rows)	rows per model: a share (0.25) or a count (50)
<code>bootstrap</code>	True	rows with replacement (bagging) or without (pasting)
<code>max_features</code>	1.0 (all columns)	columns per model: a share or a count
<code>bootstrap_features</code>	False	columns with replacement or without
<code>oob_score</code>	False	score the ensemble on out-of-bag rows (<code>oob_score_</code>)
<code>n_jobs</code>	None (1 core)	cores to use; -1 means all

- On the moons data, bagging lifts one tree from 0.856 to 0.912; with only two columns, column sampling hurts (0.648).
- On 10,000 rows, bagging lifts one tree from 0.927 to 0.945; a grid search finds 0.952.
- The OOB score (0.943) estimates test accuracy (0.945) without a test set.
- Start with bagging and `max_samples` 0.25 to 0.5; sample columns only with many columns; tune with a grid search.

Sources

- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning* (ESL), 2nd ed. Springer, section 15.2, equation 15.1.
- scikit-learn developers. Release notes, version 1.2 (December 2022), `base_estimator` renamed to `estimator`. https://scikit-learn.org/stable/whats_new/v1.2.html

7. Key terms

Term	Meaning
BaggingClassifier	scikit-learn class for bagging, pasting, random subspaces and random patches in classification
estimator	The base model that bagging copies (formerly <code>base_estimator</code>)
n_estimators	The number of base models in an ensemble

Term	Meaning
max_samples	The number or share of rows each base model gets
bootstrap	BaggingClassifier setting: draw rows with replacement (True, bagging) or without (False, pasting)
bootstrap_features	BaggingClassifier setting: draw columns with replacement or without
estimators_samples__	The row numbers each trained base model was given
estimators_features__	The column numbers each trained base model was given
verbose	scikit-learn setting that prints progress messages during training